

**Defining AI Agents: An Ontological and Topological Framework for
Organizational Systems**

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Abstract

The scalability and governance challenges of AI agents trace back to a fundamental architectural deficit: the absence of an explicit, semantically expressive, and structurally adaptive abstraction layer. The central limitation of existing management and computer science approaches is not the absence of semantic, structural, organizational, or formal methods individually, but the lack of a common abstraction layer through which these dimensions can be unified. This paper introduces an architectural framework utilizing explicit, machine-readable ontological and topological representations for AI agents to resolve this gap. Ontology specifies the semantic categories and relations constituting an AI system’s representational world, while topology models the structural organization and connectivity through which these ontological elements interact. Leveraging the formalisms of category theory, mereotopology, and graph theory, we formally define the framework and prove its five core properties: composability, interoperability, provenance preservation, auditability, and formal verifiability. Finally, we demonstrate that the implications of this framework extend far beyond pure technical representation, offering critical paradigm shifts for complex adaptive systems, AI governance, epistemic transparency, organizational theory, game theory, and composable AI ecosystems.

Keywords: ontology, topology, AI agent, multi-agent system

Defining AI Agents: An Ontological and Topological Framework for Organizational Systems

Introduction

The rapid deployment of autonomous AI agents within corporate and public infrastructures has shifted organizational focus from simple task automation to complex, distributed agent coordination. However, as institutional reliance on these systems deepens, significant systemic challenges emerge that threaten long-term scalability.

First, the complexity of multi-agent interactions are increasingly difficult to engineer, maintain, and govern (Binyamin & Ben Slama, 2022; Jennings et al., 1998; Wooldridge, 2009). Second, contemporary AI deployments often result in fragmented, implementation-specific compromises that seldom interoperate with each other, let alone build on each other (Adner, 2017; Jacobides et al., 2018). Third, the operational decisions and execution pathways of autonomous agents often lack formal predictability, necessitating frequent, if not constant, human verification (Baier & Katoen, 2008; Clarke, 2018). Fourth, AI systems remain largely alienated from their broader sociotechnical frameworks, operating without a deep integration into existing institutional contexts or organizational memory (Orlikowski, 2002; Puranam, 2021; Trist, 1981). Finally, while organizations urgently require the capability to enforce deterministic constraints on autonomous behavior, current methodologies remain limited to brittle, non-deterministic prompt engineering techniques (Dignum, 2019; Russell, 2019).

Cumulatively, these scalability and governance issues trace back to a fundamental architectural deficit: the absence of an explicit, semantically expressive, and structurally adaptive abstraction layer for AI agents. Resolving these friction points requires a paradigm shift that integrates formal semantic representation with robust structural modeling.

Existing management research has increasingly engaged with both the semantic and structural dimensions of technological systems, establishing a foundational vocabulary for

understanding autonomous agency. From a semantic perspective, scholars recognize that AI integration triggers a profound ontological shift in how organizations define, manipulate, and validate knowledge architectures (Bellesia et al., 2023; Pachidi et al., 2020). Rather than a straightforward plug-and-play adoption, the value and deployment of modern AI are deeply bounded by task architecture (Dell’Acqua et al., 2026), the operationalization of specific task environments (Felten et al., 2021), and the institutionalized organizational routines that dictate what counts as trustworthy, authoritative analysis (Anthony, 2021). Parallel to these semantic insights, a structural stream of research demonstrates that the underlying topology of communication graphs, workflows, and networks decisively shapes organizational adaptation and information processing. For instance, specific network topologies can optimize immediate fact-finding while simultaneously suppressing the diversity of theory generation (Shore et al., 2015), whereas centralized structural configurations—though typically restrictive—can paradoxically accelerate collective problem-solving and systemic flexibility during volatile environmental shifts (Bernstein et al., 2022; Cameron & Rahman, 2022). Crucially, this interplay between structure and meaning does not occur in isolation; it unfolds within decentered, relational data ecologies and broader institutional ecosystems where external stakeholders actively co-evolve and adapt to localized technological deployments (Alaimo & Kallinikos, 2021; Cui et al., 2022; Dahlke et al., 2024).

Concurrently, computer science and multi-agent systems (MAS) research (Dignum, 2009; Jennings et al., 1998; Wooldridge, 2009) has developed sophisticated approaches for agent architectures, coordination protocols, and formal verification (Baier & Katoen, 2008). While these foundational approaches offer powerful techniques for managing distributed interactions and ensuring execution correctness, the recent emergence of Large Language Model (LLM) agents has shifted attention toward dynamic orchestration frameworks, tool-use protocols, planning architectures, and prompt-based alignment mechanisms (Wang et al., 2025; Xi et al., 2023). Yet, despite these technical advances,

much of this corpus remains focused on the optimization and control of agent behavior within predefined environments. Consequently, the broader organizational, institutional, and semantic contexts within which agents operate are often treated as external assumptions rather than explicit representational objects.

Despite these advances, existing approaches remain fragmented across disciplinary and technical boundaries, severely limiting their ability to handle the realities of enterprise-scale AI deployment.

- First, current approaches lack a unified abstraction layer. While computer science focuses on micro-level execution and management theory looks at macro-level social impacts, neither offers an explicit, mathematically rigorous meta-framework that simultaneously binds semantic meaning to structural topology.
- Second, organizational-scale AI systems introduce challenges that differ qualitatively from those of isolated models. Contemporary enterprise deployments may involve hundreds of data sources, tools, services, agents, and governance requirements operating simultaneously across organizational boundaries. Without an explicit, system-wide topological map, these multi-agent systems degenerate into unmaintainable, implementation-specific entanglements.
- Third, contemporary AI challenges span organizational, computational, mathematical, and governance domains simultaneously, necessitating an interdisciplinary solution. While computer science contributes powerful methods for formal representation, verification, and system design, management research provides insights into governance, institutions, coordination, and socio-technical embeddings. Enterprise-scale AI systems increasingly require frameworks capable of integrating both.
- Finally, the future evolution of AI systems remains difficult to predict. As organizations increasingly rely on autonomous and compositional AI infrastructures,

systems must navigate unpredicted operational realities. Managing these emergent behaviors requires moving away from fragile, prompt-level patches and moving towards inherent system-level transparency, governable architectures, composability, and interoperability.

Thus, the central limitation of existing approaches is not the absence of semantic, structural, organizational, or formal methods individually, but the absence of a common abstraction layer through which these dimensions may be represented and analyzed together.

In this interdisciplinary paper, we build the framework on the dual conceptual foundations of ontology and topology, adopting compositional and verification-oriented perspectives. Ontology specifies the *semantic categories and relations* constituting an AI system’s representational world (Davis et al., 1993; Guarino, 1998), while topology specifies the *structural organization and connectivity* through which these ontological elements interact (Simon, 1962). The two are mutually constitutive. Together, these almost-intuitive concepts form the backbone of an expressive, highly adaptable, and architecturally governable framework that is nonetheless accessible for researchers as well as practitioners and regulators with diverse academic backgrounds.

Several ontological and topological concepts and their formal tools underpin this approach. Category theory informs the compositional semantics, enabling a paradigm shift from monolithic or component-based architectures to compositional AI. Mereotopology (Smith & Varzi, 1996) informs the structural and boundary relations, while graph theory (Olfati-Saber et al., 2007) models the operational topology within multi-agent systems. Leveraging these formalisms, we formally define the framework and its core components.

Rather than imposing a rigid universal schema, the framework proposes that individual multi-agent systems define a minimal common core of foundational ontological and topological dimensions required to support interoperability, governance, provenance, and system-level coordination. It thus accommodates heterogeneous representations under

shared interoperability constraints.

Nevertheless, we propose a common candidate set of dimensions for illustrative purposes and as a practical starting point. These dimensions are abstracted inductively from recurring semantic and structural patterns identified across technical standards, regulatory guidance, policy fragments, and corporate governance documents. This underlying corpus comprises the clearest official or quasi-normative formulations concerning AI agents from twenty influential organizations. In total, the candidate set includes seven ontological dimensions across four categories that are semantically fundamental, orthogonal where possible, operationally meaningful, and reusable across heterogeneous multi-agent systems. Complementing these are six topological dimensions chosen to be structurally fundamental, implementation-agnostic, compositional, operationally meaningful, and analyzable through existing formal methods.

Thanks to explicit ontological and topological representations, five core properties emerge from our proposed framework, which we formally prove. First, **composability** allows agents, workflows, and organizational structures to be combined through explicit topological relations while preserving semantic consistency under shared ontological constraints. Second, **interoperability** ensures that heterogeneous agents and systems can seamlessly exchange and interpret representations across machine-readable dimensions. Third, **provenance preservation** guarantees the explicit traceability of dependencies, transformations, and execution pathways, allowing system outcomes to be directly mapped to individual agent actions. Fourth, **auditability** emerges by externalizing semantic, operational, and provenance structures into inspectable formal representations rather than opaque implementation logic, enabling systematic examination of behavior and policy compliance. Finally, **formal verifiability** ensures the framework’s ontological and topological structures can be directly represented and analyzed through automated or semi-automated verification methods.

Crucially, the implications of our proposed framework extend beyond the formal

representations of AI agents. For complex adaptive systems, the framework provides a coherent *language* to express AI systems as adaptive organizational ecologies composed of semantically situated and structurally interoperable agents. In terms of governance, the framework transforms it from an *ex-post* procedural effort to an *ex-ante* architectural feature of the AI system. Epistemically, the framework offers system-level semantic and structural transparency as an alternative to treating AI opacity solely as an issue of machine learning models. For organizational theory, the framework establishes the view that AI systems are semantically situated sociotechnical infrastructures whose performance depends on positioning them within institutional contexts. In the realm of game theory, the framework provides the formalized strategic environment to analyze structured interactions among heterogeneous AI agents, whence *agency* arises. Finally, for composable AI ecosystems, the framework lays the basis for ecosystem-level composability, interoperability, and governance across institutional and regulatory boundaries. Taken together, our framework empowers the governance, coordination, evolution, and compositional organization of AI systems at scale.

The remainder of this paper develops as follows: We first establish the dual conceptual foundations of ontology and topology for AI agents, positioning these concepts within existing management literature. Then, we introduce the ontological–topological framework, both conceptually and formally, and prove its five core properties. This is followed by the proposal of a common candidate set of dimensions for structuring AI agents and multi-agent systems. Thereafter, we discuss the broader implications of the framework through six complementary lenses before concluding.

Conceptual Foundations

Ontology, which is the philosophical study of what exists, what matters, and how they relate to one another, has long been implicitly embedded in the management literature. For example, early works on the theory of the firm assumed firms to be rational, bounded entities (e.g. Cyert & March, 1963; Simon, 1947), open systems embedded within

environments (e.g. Katz & Kahn, 1966; Pfeffer & Salancik, 1978), or socially constructed realities (e.g. Weick, 1995); institutional theories assumed that norms, rules, classifications, and cultural scripts possess social reality and causal force (e.g. Meyer & Rowan, 1977; Thornton et al., 2012); and routines and capabilities are treated as persistent organizational entities with causal powers (e.g. Felin & Foss, 2005; Nelson & Winter, 1982; Teece et al., 1997), etc. In AI-related management research, ontology is often used implicitly through assumptions of what AI is (e.g. Shrestha et al., 2019), what agency it possesses (e.g. Murray et al., 2021), what role it has (e.g. Kellogg et al., 2020; Rahman, 2021), and how organizations relate to it (e.g. Glaser et al., 2021; Lebovitz et al., 2022).

However, explicit, machine-readable ontology that specifies the entities, relations, and constraints assumed to exist in a domain (Gruber, 1993) is inalienable from deep, repeatable, and verifiable study of AI agents and multi-agent systems (Wooldridge & Jennings, 1995).

Management scholars have likewise implicitly used topological logics to conceptualize organizations, knowledge, power, networks, innovation, platforms, and institutional fields. For example, structural holes (Burt, 1992), embeddedness (Granovetter, 1985), and interorganizational networks (Powell et al., 1996) are fundamentally topological concepts; institutional environments (e.g. DiMaggio & Powell, 1983; Scott, 2014) are structured spaces with centers, peripheries, boundaries, and regions of legitimacy; and knowledge exists within structured spaces of proximity, distance, transferability, and connectivity (Cohen & Levinthal, 1990; Lave & Wenger, 1991), etc. Indeed, contemporary research is increasingly shifting towards topological thinking with networks, ecosystems, platforms, flows, distributed agency, and fluid boundaries. AI-related management literature is also deeply topological, with increasing focus on AI's influence on the shape of organizational relations (e.g. Kellogg et al., 2020; Lebovitz et al., 2022; Murray et al., 2021; Shrestha et al., 2019).

However, explicit, machine-readable topology that specifies how the ontological

elements are organized, connected, and transformed is further required to understand, control, and manage not only multi-agent systems but also standalone AI agents within the larger organizational context.

In the conceptual and engineering framework that we propose for defining AI agents, several ontological and topological concepts provide theoretical foundations.

Category theory informs the *compositional semantics*. Our framework adopts a compositional perspective to support modular representation, compositional interoperability, and morphism-preserving transformations between agent structures and workflows. This moves the understanding of AI agents from monolithic AI¹ and component-based AI² to compositional AI³. Several applied categorical structures can be used to analyze an implementation of the framework. *Olog* is a category-theoretic schema for a domain that can be queried as databases (Spivak & Kent, 2012). *Petri net*, which is a bipartite directed graph of places and transitions with tokens that move when transitions fire, can serve as a basic model of concurrent systems with a well-defined operational semantics in the context of workflow management, where flowcharts fall short (van der Aalst, 1998). *Open Petri net* enriches Petri nets with additional compositional structure by gluing the outputs of one to the inputs of another (Baez & Master, 2020).

Mereotopology (Smith & Varzi, 1996) informs the structural and boundary relations. Our framework extends ontological specification with explicit topological relations inspired by mereotopology, enabling formal representation of part-whole structures, interaction boundaries, and contextual adjacency within multi-agent

¹ An architecture in which a single integrated model performs all functions internally without modular decomposition.

² An architecture in which the system is engineered as a collection of specialized, interoperable modules that each performs distinct functions. Component-based AIs are not always composable.

³ An architecture in which intelligence emerges through the structured combination and recombination of reusable representations, functions, reasoning steps, or modules to generate flexible and systematic behavior. Compositional AI need not be component-based, but in our framework it is.

environments.

Graph theory (Olfati-Saber et al., 2007) informs operational topology, especially in multi-agent systems. Graph-theoretical structures provide the operational representation of interaction topologies, communication pathways, dependency structures, and execution flows among agents and system components. As long as agents are relatively homogeneous and interactions reduce to edges and weights, the graph-based approach is simple and robust. Its weakness appears when one must represent local heterogeneity or constraints richer than a simple peer-to-peer coupling. Sheaf theory (Felber et al., 2025; Hanks et al., 2025) generalizes multi-agent coordination beyond graphs when agents, state spaces, or local constraints are heterogeneous; our framework is compatible but does not build on sheaves.

On these foundations, ontology and topology may be understood as conceptually distinct yet interdependent layers of an AI system. Ontology specifies the semantic categories and relations constituting an AI system’s representational world (Davis et al., 1993; Guarino, 1998), whereas topology specifies the structural organization and connectivity through which these elements interact (Simon, 1962). The two are mutually constitutive.

In the context of large language models (LLMs, Brown et al., 2020; Vaswani et al., 2017), their relevance lies not in substituting for the models themselves but in providing the semantic and structural scaffolding through which the LLM’s inputs and outputs anchor, the agent’s memory persists, its execution becomes auditable and verifiable, and multiple agents may coordinate dynamically as a coherent system.

When designed appropriately, ontological and topological structures jointly enable multi-agent systems to become more transparent, interpretable, traceable, and systematically analyzable. The ontological layer provides a shared semantic basis for system components, while the topological layer defines their patterns of interaction, composition, and information flow. Together, they establish the structural conditions

under which system behavior can be meaningfully inspected and reasoned about.

Within this setting, **provenance** (Lebo et al., 2013; Moreau, Missier, et al., 2013) provides a complementary mechanism for auditability by enabling the traceability of how agents, tools, data, and intermediate decisions contribute to observed outputs. In contrast, **formal verification** (Lamport, 1998; Pnueli, 1977) provides verifiability by establishing mathematical guarantees that system behaviors satisfy explicit specifications.

In open, self-organizing multi-agent systems, these concerns motivate a preference for compositional architectures and reusable local proofs, which allow auditability and verifiability to scale with system complexity.

The Ontological—Topological Framework of AI Agents

Building on the ontological and topological foundations as well as the compositional and verification-oriented perspectives established in the preceding sections, the proposed framework formalizes AI agents and multi-agent systems through explicit ontological and topological dimensions expressed in machine-readable form.

AI agents within the framework are defined relative to an explicit domain ontology that establishes the foundational semantic environment. This context defines the core entities, roles, policies, constraints, and organizational semantics within which agents operate and through which ontological and topological dimensions acquire meaning. By externalizing these system-level semantics from individual agent definitions, the framework supports shared governance, interoperability, provenance consistency, and system-wide verification across heterogeneous agents and workflows.

Ontological dimensions define the semantic characteristics, capabilities, identities, constraints, and contextual meanings associated with agents, whereas topological dimensions define the structural, relational, operational, and compositional organization through which agents interact within the multi-agent system.

Rather than imposing a universal schema, the framework proposes that individual multi-agent systems define a minimal common core consisting of foundational ontological

and topological dimensions required to support interoperability, governance, provenance, and system-level coordination within a given operational context. The framework therefore accommodates heterogeneous ontological and topological representations under shared interoperability constraints.

Because these dimensions are represented explicitly and formally, agent specifications are inherently machine-readable and amenable to automated composition, orchestration, provenance tracking, auditing, and formal verification. Compositionality emerges from explicitly defined topological relations rather than from hardcoded architectural hierarchies or implementation-specific orchestration mechanisms. Existing formal tools derived from category theory, mereotopology, and graph theory, etc. may therefore be applied to analyze the structural consistency, compositional validity, provenance relations, and operational semantics of AI agents and the multi-agent systems.

Figure 1 illustrates the conceptual architecture of the proposed framework, including the relationships among the domain ontology, agent ontology, topology, and the minimal common core, as well as their interfaces with provenance and formal verification through explicit machine-readable representations. Figure 2 presents an example topological graph representation of an individual agent situated within a multi-agent system.

Under our proposed framework, neither the system context nor the operational structure of the AI agent is treated as a black box, even when the underlying large language model remains partially opaque. The framework instead externalizes semantic, structural, and operational characteristics into explicit ontological and topological representations that are amenable to inspection, governance, auditing, provenance analysis, and formal verification.

The **compositional semantics** of the framework arise from the explicit representation of ontological and topological dimensions and the formal relations defined among them. Agents may therefore be composed, decomposed, extended, or orchestrated through topological relations while preserving semantic consistency under the constraints

of the domain ontology and the minimal common core. Because compositional structure is represented explicitly rather than embedded implicitly within implementation-specific logic, heterogeneous agents and workflows may interoperate through shared relational semantics and machine-readable interfaces. This compositional perspective is compatible with existing formal approaches derived from category theory and Petri nets, enabling modular reasoning about agent composition, transformation, orchestration, and dependency structures within multi-agent systems.

The **operational semantics** of the framework emerge from the topological organization of agents, workflows, dependencies, and interaction pathways within the multi-agent system. Topological relations may therefore represent execution sequences, coordination boundaries, information flows, containment structures, or dependency constraints, enabling agent interactions and system behavior to be analyzed as explicit operational structures rather than opaque runtime processes.

Thanks to explicit, machine-readable ontological and topological representations, provenance relations, execution histories, dependency chains, and policy constraints may be systematically externalized and analyzed. The framework therefore enables auditing and formal verification not merely at the level of isolated agents, but across composed workflows and organizationally situated multi-agent systems.

Under these conditions, the framework is intended to satisfy several core properties:

1. *Composability* is the ability of agents, workflows, and organizational structures to be combined through explicit topological relations while preserving semantic consistency under shared ontological constraints.
2. *Interoperability* is the capacity of heterogeneous agents and systems with a shared minimal common core to exchange and interpret representations through machine-readable ontological and topological dimensions.
3. *Provenance preservation* refers to the explicit traceability of dependencies,

transformations, interactions, and execution pathways across the multi-agent system, including the ability to relate system outcomes to the actions and decisions of individual agents or coordinated agent workflows.

4. *Auditability* emerges from the externalization of semantic, operational, and provenance structures into inspectable formal representations rather than opaque implementation logic, thereby enabling systematic examination of agent behavior, decision pathways, policy compliance, and organizational processes.
5. *Formal verifiability* refers to the ability of the framework’s ontological and topological structures to be represented and analyzed through formal methods amenable to automated or semi-automated verification techniques.

Below we characterize the framework formally through explicit definitions of ontological representation and topological composition. To help readers with varying mathematical backgrounds follow the rest of this section, we explain in plain terms every mathematical word used in Appendix A. The five core properties are also stated, while detailed mathematical representations and proofs are provided in Appendix B.

Conceptually, keeping ontology and topology separate lets us reason about meaning and structure independently. Formally, both live in the language of typed graphs.

Definition 1 (Graphs and typed graphs). A *directed multigraph* is $G = (V_G, E_G, s_G, t_G)$ with source and target maps $s_G, t_G : E_G \rightarrow V_G$. A morphism $f : G \rightarrow H$ is a pair (f_V, f_E) with $s_H \circ f_E = f_V \circ s_G$ and $t_H \circ f_E = f_V \circ t_G$. Graphs and morphisms form the category **Grph**, which is a presheaf topos: it is complete, cocomplete, and adhesive (monomorphisms are stable under pushout). For a fixed object T , the slice category **Grph**/ T has objects (G, τ_G) with a typing $\tau_G : G \rightarrow T$, and morphisms commuting with the typings. The forgetful functor **Grph**/ $T \rightarrow \mathbf{Grph}$ creates colimits, so **Grph**/ T is cocomplete (pushouts computed in **Grph**) and, as a slice of a topos, adhesive.

These three facts—(1) cocompleteness, (2) pushouts are computed by gluing, and (3) adhesivity (gluings keep distinct things distinct)—are the only categorical inputs used in the formalization below.

Before describing any single agent, we must fix the shared vocabulary and rules that everything must respect. This is expressed by the domain ontology and the agent’s conformance.

Definition 2 (Domain ontology). A *domain ontology* is

$$O = (T_O, \sqsubseteq_O, \Phi_O)$$

where T_O is a finite *type graph* (vertices are sorts such as **Agent**, **Tool**, **Data**, **Human**; edges are relation symbols with fixed source and target sorts, e.g. $invoke : \mathbf{Agent} \rightarrow \mathbf{Tool}$, $escalate : \mathbf{Agent} \rightarrow \mathbf{Human}$); $\sqsubseteq_O$ is a partial order on sorts (the is-a hierarchy); and Φ_O is a finite set of *negative constraints* $\neg P$, each given by a finite connected typed pattern $P \in \mathbf{Grph}/T_O$.

Definition 3 (Conformance). A *typed structure* over O is an object (G, τ_G) of $\mathbf{Grph}/T_O$ whose typing respects $\sqsubseteq_O$. It *conforms*, written $G \models O$, iff for every $\neg P \in \Phi_O$ there is no typed homomorphism $P \rightarrow G$. The full subcategory of conformant structures is $\mathbf{Str}(O)$.

“Conformant” is the precise word for “this setup obeys the specifications.” Almost everything below is about keeping setups conformant as we build and combine them. Stating rules as forbidden patterns is what makes them easy to check and, crucially, easy to combine.

Definition 4 (Agent). Relative to O , an *agent* is a tuple

$$a = (Ont_a, Top_a)$$

where Ont_a is the set of ontological vertices with typing into T_O , while Top_a is the set of topological vertices; they represent the ontological and topological dimensions of the agent.

The set of policies governing a , $\Pi_a \subseteq \Phi_O$, and the set of states (memory), Σ_a , are subsets of Ont_a . The *boundary graph*, B_a , which is a finite T_O -typed graph whose edges are the interface relations, is a subset of Top_a . The agent is *well-formed* iff $B_a \models O$.

To talk about boundaries cleanly we need two everyday relations: “part of” and “in contact with”. The point where an agent’s parts touch the outside is its interaction boundary; we cannot hide a connection by tucking a part inside a larger unit, as it still shows at the larger unit’s boundary.

Definition 5 (Mereotopological boundary). Let U be the set of parts, with parthood P (a partial order) and connection C (reflexive, symmetric). Define overlap $\text{Ov}(x, y) :\Leftrightarrow \exists z(P(z, x) \wedge P(z, y))$ and external connection $\text{EC}(x, y) :\Leftrightarrow C(x, y) \wedge \neg \text{Ov}(x, y)$, with the monotonicity axiom $P(x, y) \Rightarrow \forall z(C(z, x) \Rightarrow C(z, y))$. The *interaction boundary* of a in environment e is $\partial(a, e) = \{(x, y) : P(x, a) \wedge P(y, e) \wedge \text{EC}(x, y)\}$. The edges of B_a are exactly these external-contact links.

An implication is *Boundary monotonicity*, which says that, if an agent is part of an agent system, then every external connection of the agent is inherited by the system. The formalization and proof can be found in Appendix B as Proposition 1.

Different agents need not agree on everything; however, they need to agree on a small shared part—the words used whenever they interact across a boundary. This is the minimal common core. On top of it, each party keeps its own richer vocabulary.

Definition 6 (Minimal common core). A *minimal common core* is a designated subgraph $K \hookrightarrow T_O$. For agents over distinct ontologies O_i , a *core embedding* is a monomorphism $\kappa_i : K \hookrightarrow T_{O_i}$. The core carries the inter-agent vocabulary.

Definition 7 (Multi-agent system and gluing). A *multi-agent system* is

$$M = (O, K, \{a_i\}, G)$$

where G , the system topology, is the colimit (gluing) of the boundary graphs B_{a_i} along the shared interfaces of every cross-agent interface being typed within K . An *interface* between

$A, B \in \mathbf{Grph}/T_O$ is a typed graph I with monos $I \hookrightarrow A$ and $I \hookrightarrow B$; the *gluing* $A \oplus_I B$ is their pushout. Split $\Phi_O = \Phi_{\text{loc}} \sqcup \Phi_{\text{core}}$, where patterns in Φ_{loc} use no core relation type; the gluing is *core-guarded* when it verifies every $\neg P \in \Phi_{\text{core}}$ on the seam I at glue time.

If two parts each obey the rulebook, and we join them correctly along a shared connector, the result still obeys the rulebook. So we can build and grow systems by composition without the join secretly introducing a violation. This is the property of **Composability**. It is formalized as Theorem 1 in Appendix B. Figure 3 shows as an example the gluing of two conformant pieces A, B along a shared interface I (here a common Data node).

If different parties share the common-core vocabulary, anything written in that core means the same to all of them, and the two systems can be merged without their meanings clashing or collapsing. This is the property of **Interoperability**. It is formalized as Theorem 2 in Appendix B. In plain words, merging glossaries that share a common part never forces two different terms to become one, and anything written purely in the shared core lines up identically for everyone.

Definition 8 (Provenance graph and provenance functor). A *provenance graph* is a finite labelled directed acyclic graph (DAG) with node types **Entity**, **Activity**, **Agent** and edges orienting effects to cause

$$\left\{ \begin{array}{l} \text{used}(\text{Act} \rightarrow \text{Ent}) \\ \text{wasGeneratedBy}(\text{Ent} \rightarrow \text{Act}) \\ \text{wasDerivedFrom}(\text{Ent} \rightarrow \text{Ent}) \\ \text{wasAssociatedWith}(\text{Act} \rightarrow \text{Agent}) \end{array} \right.$$

It is acyclic iff it admits a topological order. Let **Exec** be the category of runs (composed by concatenation at matching cuts) and **Prov** that of provenance graphs (composed by gluing outputs to inputs). The *provenance functor* $P : \mathbf{Exec} \rightarrow \mathbf{Prov}$ emits, per step, provenance graphs that direct outputs to activities, entities, agents, and inputs.

In the framework, every output carries an unbroken record of what produced it—which agent acted, what it used, what it was derived from—and that record never loops. When the system runs step after step, the records simply chain together, and each part’s history is read off from the whole. This is the property of **Provenance preservation**. It is formalized as Theorem 3 in Appendix B. In plain words, with every step timestamped, every “caused-by” arrow points back in time: a loop would put something before itself, which is impossible.

Definition 9 (Runtime Externalization). For a run r and morphism $\alpha : \text{Runs} \rightarrow E$, its externalization function is

$$r \mapsto \text{Ext}(r) = (T_t, G_t, P(r))$$

where T_t is the typing of touched structures, G_t is the topological trace, and $P(r)$ is the provenance functor. A property $\psi : \text{Runs} \rightarrow \{0, 1\}$ is *auditable* iff it factors through α , i.e. $\psi = \bar{\psi} \circ \alpha$, equivalently iff ψ is constant on each fibre $\alpha^{-1}(e)$.

Any question that an auditor would ask—did this follow policy, who was responsible, what was this based on—can be answered purely from the written record without seeing inside any agent’s reasoning; additionally, audits of the parts combine into an audit of the whole. This leads to the property of **Auditability**. It is formalized as Theorem 4 in Appendix B.

Definition 10 (Safety property and contracts in operation). The semantics of M is a transition system $T_M = (\Sigma, \Sigma_0, \rightarrow)$ over the operational semantics Λ , which is the disjoint union of the operational subsets of $\{Ont_i\}$ and $\{Top_i\}$. Opacity is nondeterminism inside $\rightarrow$.

(1) A *safety property* φ is a prefix-closed set of finite traces; $T \models \varphi$ iff all finite traces lie in φ .

(2) A *contract* $\langle \text{Assumption}, \text{Guarantee} \rangle$ is a pair of prefix-closed trace sets; $M \models \langle \text{Assumption}, \text{Guarantee} \rangle$ iff for every trace σ , whenever all proper prefixes of σ lie

in *Assumption* then $\sigma \in \textit{Guarantee}$.

Finally, safety guarantees can be established for each component on its own, using only its visible behavior, then combined into a guarantee for the whole, without redoing the proof for the combination. This is the property of **Verifiability**. It is formalized as Theorem 5 in Appendix B.

Ontological and Topological Dimensions of AI Agents

While the framework does not prescribe a fixed set of ontological or topological dimensions, we propose a common candidate set. They are abstracted inductively from recurring semantic and structural patterns identified across technical standards, regulatory guidance, policy fragments, and corporate governance documents published by corporations, regulators, and standard-setting organizations.

The corpus comprises the clearest official or quasi-normative formulations concerning AI agents from twenty influential organizations. These sources converge on the view that an AI agent is a goal-oriented software system, supported by one or more models, that selects actions, uses tools, and manages enough state and context to execute multi-step tasks with some degree of autonomy. However, the sources diverge on the boundaries between assistants, workflows, agents, and multi-agent systems, reflecting the differing institutional perspectives of the three constituencies: vendors approach AI agents as users of their products, regulators as risks requiring governance, and standards bodies as interoperable technical subjects.

In synthesizing the dimensions, we recommend having a minimal common ontological and topological core sufficient for interoperability and regulatory consistency. Without such a shared core, systems cannot be meaningfully compared, obligations cannot be calibrated, and standardized interfaces cannot be designed. At the same time, an overly rich universal definition would be counterproductive. Instead, with ontological profiling dimensions and topological structuring, the framework can accommodate the diversity of individual AI agents and multi-agent systems.

In total, the candidate set includes seven ontological dimensions in four categories. They are semantically fundamental, orthogonal where possible, operationally meaningful, and reusable across heterogeneous multi-agent systems. Table 2 offers an overview with some example elements to help readers form a more intuitive understanding of the semantics of agent ontology.

1. **Semantic identity** specifies *what the agent is* in two dimensions. *Identity* characterizes the AI agent’s classification, versioning, lifecycle, and agent type; it supports traceability, provenance attribution, and verification-oriented reasoning. *Role* captures the agent’s purpose, organizational function, responsibility, and governance context; it is crucial in organizational multi-agent systems.
2. **Functional semantics** specifies *what the agent does* by defining its *Capability*, which is the agent’s functional repertoire. It includes the AI agent’s functional competencies, available actions, supported operations, and reasoning abilities, thus central to composition and orchestration.
3. **Contextual semantics** specifies *what the agent knows* in two dimensions. *Knowledge* delineates data accessibility, domain specialization, contextual grounding, and model scope; it situates the agent in the organizational, system, and knowledge context. *State* defines the AI agent’s memory, situating the agent in the workflow / session context; it may also be considered operational or runtime semantics.
4. **Governance semantics** specifies *what constraints the agent* in two dimensions. *Constraint* governs the AI agent’s permissions, restrictions, obligations, compliance, and safety boundaries; it is crucial for formal verification. *Artefacts* defines the operational artefacts that the agent produces for governance, traceability, and verifiability; it may be considered operational or runtime semantics, too.

State and Artefacts occupy a boundary position between ontological representation

and operational semantics because they capture runtime contextualization and externally inspectable operational traces.

The candidate set also includes six topological dimensions chosen to be structurally fundamental, implementation-agnostic, compositional, operationally meaningful, and analyzable through existing formal methods. Table 3 offers an overview with some example elements as before.

1. *Connectivity* is the most fundamental topological dimension. It defines the admissible interaction and communication relations among agents and system components, including which agents or agent types can interact, communication pathways, and information exchange structure, etc. Connectivity is the core graph-theoretic layer.
2. *Containment* is crucial for governance and provenance. It defines the membership relations, including subsystem membership, organizational nesting, bounded contexts, swarms, and domains of operation, etc. Containment is where mereotopology becomes important.
3. *Dependency* is essential for provenance, orchestration, auditability, and verifiability. It defines workflow-, system-, and organizational-level dependencies, including execution dependencies, data dependencies, authorization dependencies, resource dependencies, and prerequisite structures, etc.
4. *Coordination* is where operational semantics becomes explicit. It defines the control relations in the multi-agent system, including orchestration structure, delegation, supervision, synchronization, and control flow, etc.
5. *Boundary* is vital for governance, auditability, and compliance. It defines the trust and policy boundaries, security zones, organizational limits, and jurisdictional partitions. Boundary is often neglected but contributes significantly to the externalization and governance of agent behavior and operational context.

6. *Temporal* is workflow-oriented and elective. It defines the execution relations, including sequencing, concurrency, synchronization, lifecycle transitions, and state progression, etc. The Temporal dimension aligns strongly with Petri nets and operational semantics.

Discussion

The proposed ontological–topological framework (the framework henceforth) has implications extending beyond the formal representation of AI agents. By externalizing the semantic and structural characteristics of AI systems and positioning them in the operational and organizational contexts through explicit, machine-readable ontological and topological dimensions, the framework enables AI systems to be analyzed as compositional, governable, strategically interactive, and organizationally situated structures rather than isolated computational artefacts. This shift has important implications for the study of adaptive AI systems, AI governance, organizational theory, AI strategy, and strategic interaction among autonomous or semi-autonomous agents.

In the following discussion, we examine the framework through multiple complementary lenses, including complex adaptive systems, governance and formal methods, epistemic transparency, organizational theory, game theory, and composable AI ecosystems.

AI Systems as Compositional and Adaptive Structures

Viewed through the lens of complex adaptive systems (Holland, 1995; Mitchell, 2009), the framework provides the language to understand AI systems as dynamically composable organizational structures whose behavior emerges from explicit ontological and topological relations among agents. Ontology describes the semantic categories and relations, while topology describes the structural organization and connectivity. The same entity, such as an AI model or a database, may be represented through multiple elements, both ontological and topological: The emphasis is not on enforcing a single canonical representation but on providing a coherent representational *language* through which the

same idea may be expressed in multiple semantically meaningful ways, each with its own nuances, while preserving the five core properties of the framework.

Because agents are represented through interoperable semantic and structural dimensions, heterogeneous agents may be composed, reorganized, extended, or coordinated without requiring rigid implementation-level coupling (Wooldridge, 2009). As organizational objectives and environmental conditions evolve, these changes can be reflected in the ontological constraints and topological relations, allowing the AI system to adapt and/or reconfigure structurally without major overhauls.

Within such systems, autonomy is not treated as an isolated intrinsic property of individual agents but as an operational condition shaped by organizational topology, dependency structures, governance constraints, and coordination mechanisms, etc. Consequently, the framework provides a basis for analyzing AI autonomy (Humberd & Latham, 2026) as well as forms of algorithmic management (Kellogg et al., 2020; Stark & Vanden Broeck, 2024) in which authority, supervision, delegation, and decision-making are distributed dynamically across human and AI organizational structures.

More broadly, the framework breathes life into the view that AI systems may increasingly resemble adaptive organizational ecologies composed of semantically situated and structurally interoperable agents rather than monolithic standalone models. This perspective shifts attention from isolated model capabilities towards the governance, coordination, evolution, and compositional organization of AI systems at scale.

AI Governance with Formal Methods

The framework empowers the governance of AI systems with explicit machine-readable semantic and structural representations—this transforms governance from an *ex-post* procedural effort (Grote et al., 2024; Raisch & Krakowski, 2021) to an *ex-ante* architectural feature (Floridi et al., 2018). Indeed, the main AI governance challenges reside in the difficulty of tracing, interpreting, auditing, and verifying the behavior of increasingly complex and distributed AI infrastructures, particularly in

multi-agent systems where responsibilities, interactions, and operational dependencies are dispersed. By externalizing the semantics and structures of AI systems, the framework provides a formalized substrate through which governance structures may be embedded directly into the representation and composition of the AI systems themselves, thereby addressing many governance challenges structurally rather than relying solely on external policy controls imposed upon an opaque computational system (Kim et al., 2024).

Within the framework, provenance emerges naturally from explicit ontological and topological representations: Because constraints, dependencies, coordination structures, execution pathways, and operational artefacts are represented formally, system outcomes can be traced to the actions, decisions, and interactions of individual agents. In this context, provenance extends beyond simple execution logging to include semantic traceability across the multi-agent system. These provenance relations in turn support auditability by externalizing operational and organizational structures into inspectable representations as an architectural feature of the system rather than *ad hoc* patchwork of implementation-specific logics.

Further, the framework supports the application of formal verification methods to AI systems—not only to isolated agent behavior but also at the organizational and workflow levels to dependency structures, coordination constraints, provenance relations, governance policies, and system-wide operational semantics. Existing formal methods derived from category theory, graph theory, provenance, and formal verification may therefore be leveraged to analyze policy consistency, dependency integrity, compositional correctness, and coordination behavior across heterogeneous multi-agent systems.

Crucially, these architectural governance features are partly the result of composability; consequently, governance is not compromised at the organizational, system, or workflow levels as the multi-agent system scales up or down. The framework thus preserves governance structures as AI systems evolve and breathes legitimacy into multi-agent systems in the organizational context (Dignum, 2019; Weibel et al., 2025).

Overall, this analysis points to a shift from reactive governance towards governance-oriented AI architectures in which organizational constraints, provenance, auditability, and verification are *embedded structural features* of the AI system. This perspective becomes increasingly important as AI systems evolve towards distributed, interoperable, and semi-autonomous organizational ecologies spanning institutional and regulatory boundaries.

Semantic and Structural Transparency in AI Systems

Epistemically, the framework offers system-level transparency as an alternative to treating AI opacity solely as an issue of machine learning models (Burrell, 2016). While advanced AI models such as large language models may remain partially opaque at the parametric level, sufficiently explicit semantic, structural, operational, and organizational representations at the system level (Ananny & Crawford, 2018), especially in distributed and compositional multi-agent structures, can nevertheless help us understand, predict, and control system behavior within broader organizational workflows.

By externalizing ontological and topological dimensions, the framework turns agent roles, capabilities, coordination structures, governance constraints, interaction pathways, and organizational contexts, etc. into formal, inspectable representations rather than implicitly embedding them within an opaque implementation logic. Thus, AI systems can be analyzed not merely as isolated predictive models but as semantically situated organizational structures whose operational relations and governance mechanisms are explicitly represented. That is, instead of treating opacity exclusively as a model-level problem (Rahman, 2021), agents, workflows, AI system, and organizational contexts can themselves be rendered transparent, even if parts of the underlying model remain opaque. The framework therefore shifts emphasis from the interpretability of isolated model internals towards operational intelligibility of AI systems as organizationally situated and compositional structures.

Crucially, system-level transparency empowers organizations in AI adoption by

bolstering trust (van den Broek et al., 2025; Vanneste & Puranam, 2024) besides addressing the issues of interpretability and governance. It also affords organizations the means to create traceable, auditable, and verifiable practices surrounding AI use (Hopf et al., 2023; Lebovitz et al., 2022).

Organizationally Situated AI

The framework reinforces the view that AI agents cannot be understood as context-free computational entities independent of organizational semantics and structures: AI systems derive meaning, legitimacy, operational constraints, and functional responsibilities from the organizational contexts within which they are deployed. As AI becomes ever more integrated into organizational workflows, decision-making processes, and coordination structures, the effectiveness of AI initiatives depends not only on model performance but also increasingly on the explicit representation of the organizational contexts (Krakowski et al., 2023).

Within the framework, domain ontology provides the foundational semantic environment through which organizational entities, governance structures, responsibilities, workflows, and institutional constraints, etc. are represented explicitly. Rather than an isolated representation detached from operational realities, agent ontology therefore *derives meaning relationally* from the broader organizational context. In this perspective, organizational context is not external metadata surrounding AI systems but part of the semantic substrate through which AI agents acquire operational meaning and legitimacy. Consequently, AI agents can be situated within explicit organizational structures that define authority relations, delegation mechanisms, compliance requirements, operational boundaries, and coordination responsibilities, etc. across both human and AI participants (Krakowski et al., 2025; Puranam, 2021; Stelmaszak et al., 2026). This grows in importance as organizations adopt distributed multi-agent systems spanning departments, workflows, platforms, institutional boundaries, and external ecosystems. Explicit semantic and structural representations also reduce coordination ambiguity, improve operational

consistency, support organizational memory, and facilitate the scalable integration of AI agents into evolving organizational processes.

More broadly, the framework pushes the view that AI systems in organizations may increasingly resemble semantically situated sociotechnical infrastructures rather than collections of isolated tools (Anthony et al., 2023). Organizational AI performance therefore increasingly depends on the explicit representation, governance, coordination, and compositional organization of AI agents within institutional contexts (Einola & Khoreva, 2023).

Strategic interaction and the Agency of AI Agents

Viewed through a game-theoretic lens, the framework provides a formalized substrate to understand increasingly autonomous and compositional AI systems as strategic environments of structured interaction among heterogeneous agents operating under explicit semantic, organizational, and operational constraints. As multi-agent systems evolve towards distributed organizational ecologies involving coordination, delegation, supervision, competition, and cooperation, AI behavior, notably *agency*, emerges relationally from interactions among agents.

Meaningful game-theoretic analysis of AI systems requires explicit representations of agent roles, dependencies, authority relations, coordination mechanisms, and operational constraints (Nisan et al., 2007). The externalization of ontological and topological dimensions by the framework delivers such structures that otherwise are often only implicitly embedded within the implementation logic. In the framework, connectivity, dependency, coordination, boundary, and containment relations, etc. collectively define the interaction topology within which agents exchange information, compete for resources, coordinate actions, delegate responsibilities, or operate under asymmetric authority structures. In this perspective, organizational workflows, governance constraints, and operational dependencies may themselves be interpreted as components of a broader strategic environment shaping agent behavior.

Crucially, agency is therefore not treated as an isolated intrinsic property of individual agents but as a relational condition shaped by incentives, dependencies, coordination structures, governance constraints, and organizational topology (Jensen & Meckling, 1976; Simon, 1972). Agent behavior may consequently be analyzed in terms of strategic adaptation, delegation dynamics, coordination equilibria, authority asymmetries, and institutional constraints, etc. operating across both human and AI organizational participants (Puranam, 2021). Thus, the framework also creates opportunities for richer analyses of cooperation, competition, coordination, incentive alignment, and governance within large-scale multi-agent systems (Shoham & Leyton-Brown, 2008).

Towards Organizational AI Ecosystems

An additional perspective that the framework unlocks is the emergence of composable AI ecosystems (Adner, 2017; Jacobides et al., 2018) with heterogeneous agents spanning institutional, organizational, and regulatory boundaries as a new driver of AI adoption. Within such ecosystems, composability and interoperability are foundational architectural principles. Explicit ontological and topological representations allow agents to be discovered, composed, orchestrated, audited, verified, and reconfigured without tight implementation-level coupling. This becomes increasingly valuable as organizations integrate internally developed agents with external services, foundation models, digital platforms, and autonomous organizational processes, etc. Rather than constructing monolithic AI architectures, organizations may increasingly assemble dynamic configurations of interoperable agents whose capabilities, responsibilities, and relations remain explicitly represented.

Likewise, the same explicit semantic and structural foundations provide a basis for ecosystem-level governance. Provenance, auditability, and verification would become exceedingly difficult when AI systems span conventional boundaries. However, by externalizing ontological and topological dimensions, the framework provides mechanisms through which governance may extend beyond individual agents and organizations towards

larger networks of interacting AI systems.

In this vision, AI systems are evolving sociotechnical infrastructures composed of semantically situated and structurally interoperable agents. They may require analytical approaches that integrate organizational theory, governance, formal methods, strategic interaction, and systems engineering. The framework therefore contributes not only a representational approach for defining AI agents but also a conceptual foundation for studying and governing composable AI ecosystems.

Conclusion

This paper has introduced a formal, interdisciplinary framework that utilizes explicit, machine-readable ontological and topological representations to address the fundamental architectural deficits hindering enterprise-scale AI deployment. Within this framework, AI agents are defined relative to an explicit domain ontology that establishes their foundational semantic environment. Ontological dimensions capture semantic characteristics, capabilities, constraints, and contextual meanings, whereas topological dimensions capture the structural, operational, and compositional relations through which agents interact. Leveraging the formalisms of category theory, mereotopology, and graph theory, we have established a mathematically rigorous abstraction layer that simultaneously binds the semantic representations of ontology with the structural organization of topology. Through this approach, we formally established five foundational properties essential for scalable AI systems: composability, interoperability, provenance preservation, auditability, and formal verifiability. Rather than enforcing a rigid, universal schema, this framework works with a flexible common core capable of accommodating heterogeneous agent architectures under shared interoperability constraints.

The implications of this work extend far beyond technical representation. By binding semantic meaning to structural topology, we offer a crucial paradigm shift for how autonomous systems are embedded within institutional settings. Thanks to explicit ontological and topological representations, the framework provides the *language* to express

AI systems as semantically situated and structurally interoperable adaptive organizational ecologies. This in turn moves the discourse surrounding AI governance away from fragile *ex-post* algorithmic interventions and towards deterministic *ex-ante* architectural designs. Within this paradigm, system-level semantic and structural transparency serves as a definitive guardrail against model opacity. Furthermore, organizational performance is optimized by positioning these systems as sociotechnical infrastructures directly within prevailing institutional contexts, while strategic agency emerges naturally from structured interactions within formalized environments. Crucially, the framework provides a shared language that allows computer science to explicitly represent institutional context, while equipping organizational theory with the formal mathematical methods required to enforce provable boundaries on autonomous agency.

While this paper establishes these formal foundations, several horizons for future research remain. First, empirical mapping of the proposed common candidate set of dimensions in specific enterprise environments will identify how organizations can best design their localized common core. Second, empirical validation is required to study the runtime overhead and operational dynamics of this framework within dense enterprise infrastructures. Third, empirical studies focusing on the institutional dynamics that emerge when AI systems built on this framework are deployed may reveal unexamined cultural and regulatory shifts. Fourth, future research may investigate how ontological and topological representations evolve over time as AI systems adapt, reorganize, and interact across organizational boundaries. Finally, future technical work can focus on compiling ontological and topological dimensions into automated or semi-automated formal verification pipelines. By transitioning AI engineering from implementation-specific compromises to composable, governable architectures, this framework lays the foundation for organizing, coordinating, and evolving compositional AI ecosystems at scale.

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Table 1*Topological concepts and management equivalents.*

Topological Concept	Management Equivalent
Node	Firm, actor, team
Edge	Relationship, alliance, workflow
Cluster	Strategic group, community
Centrality	Power, influence
Structural hole	Brokerage opportunity
Boundary	Organizational interface
Connectivity	Coordination capacity
Distance	Cognitive or institutional gap
Embeddedness	Relational dependence
Modularity	Organizational decomposition
Flow	Information or resource movement
Periphery	Marginal organizational position
Folding	Reorganization / recombination
Emergence	Collective organizational behavior

Table 2

The set of candidate ontological dimensions.

Category	Dimension	Example elements
Semantic identity	Identity	version, persistence, uniqueness, LLM, embeddedness
	Role	delegate, supervisor, orchestrator, verifier, coordinator
Functional semantics	Capability	code execution, physical actuation, tool use
Contextual semantics	Knowledge	data access, sensor use, Web access
	State	RAG ¹ , shared multi-agent memory
Governance semantics	Constraint	heuristics, business policies, budgets, kill switch
	Artefacts	logs, metrics, replay, registry, incident response

¹ RAG (Retrieval-Augmented Generation) in ontology is an advanced method where an AI's responses are anchored not just to retrieved text documents, but to a formal, structured semantic model of the data (Sharma et al., 2025).

Table 3*The set of candidate topological dimensions.*

Dimension	Purpose	Example elements
Connectivity	Connectivity, communication	Peer-to-peer, hierarchical, mesh, hub-and-spoke
Containment	Structure	agent belongs to workflow, workflow belongs to department, sub-agent inside orchestration structure
Dependency	Operational prerequisites	Agent A depends on the output of Agent B, verification depends on retrieval
Coordination	Control, orchestration	voting, arbitration, consensus structure
Boundary	Trust, governance	internal vs external agents, human vs autonomous domains, regulated data boundary
Temporal	Execution flow, state progression	sequential workflow, concurrent agent collaboration, staged orchestration

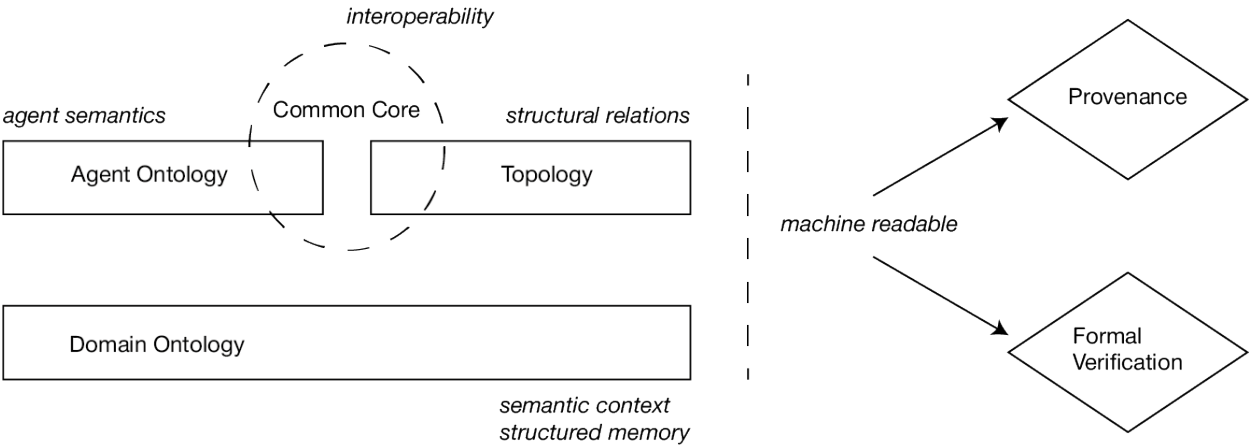
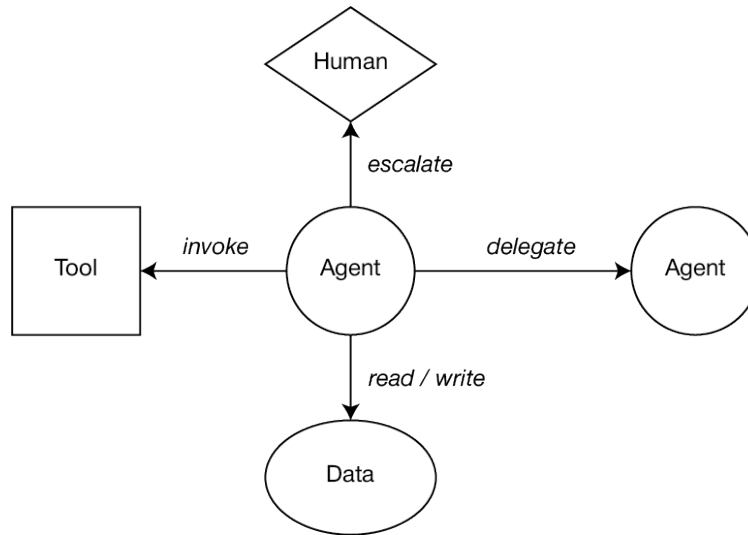
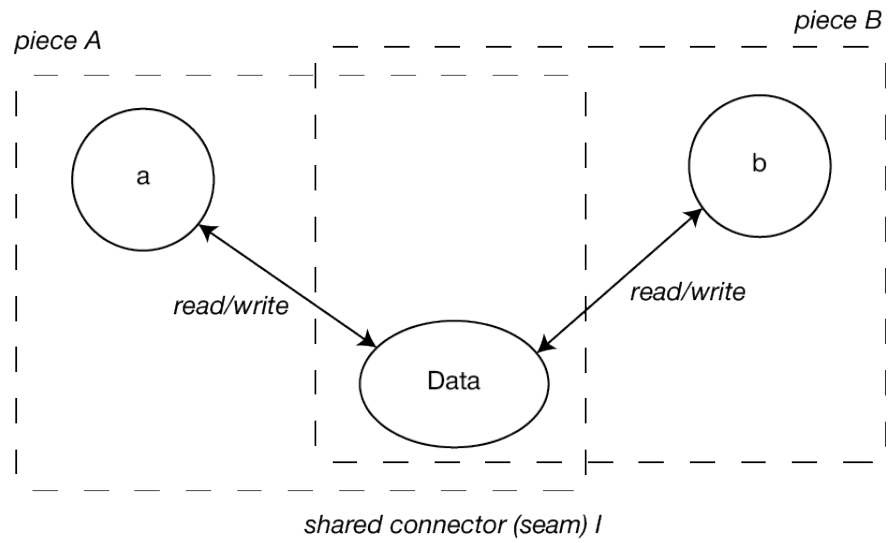


Figure 1
The ontological-topological framework of AI agents.

**Figure 2**

Simple graph of an agent.

**Figure 3**

Gluing pieces along a seam.

Appendix A

Mathematical Vocabulary

This section explains every mathematical word used in the formalization of the framework, focusing on the intuition.

Set, element, function. A *set* is a collection of its *elements*. A *function* $f : X \rightarrow Y$ is a rule assigning to each thing in X exactly one thing in Y .

Relation. A *relation* between X and Y pairs up some elements of X with some of Y , with no “exactly one” requirement. A function is the special case of a relation where every input has exactly one output.

Directed multigraph. A *graph* is a set of *nodes* (the dots) together with *edges* (the arrows) between them. “Directed” means each edge has a direction, running *from* one node *to* another. “Multi” means we allow several distinct edges between the same two nodes. Formally a directed multigraph is a set of nodes, a set of edges, and two functions—*source* and *target*—stating where each edge starts and ends.

Morphism (structure-preserving map). A *morphism* is a map from one structure to another that respects the structure. For graphs, a morphism sends nodes to nodes and edges to edges in a way that never tears an arrow loose: if an edge runs from u to v , its image must run from the image of u to the image of v . Think of overlaying a small org chart onto a bigger one so that every reporting line still lands correctly.

Monomorphism (“mono”, an inclusion). A *monomorphism* is a morphism that keeps things distinct—it never merges two different elements into one. The mental picture is a sub-structure sitting inside a larger one: a small graph embedded in a big one without folding any two nodes together. We draw it as $\hookrightarrow$. This “no merging” property guarantees that joining pieces does not accidentally collapse two distinct things.

Category. A *category* is a collection of objects together with the morphisms between them, where morphisms can be *composed* (do one, then the next) and each object has an identity morphism (do nothing). The category we use, written **Grph**, has graphs as

its objects and graph morphisms as its arrows. Categories let us state facts about *all* structures of a kind at once, rather than one structure at a time.

Limit and colimit (the two ways to combine structures). Category theory has two universal ways of combining objects. A *colimit* *glues things together*; it merges structures along shared parts. A *limit* does the opposite, carving out the part common to several structures (a synchronization or intersection). We only invoke colimits, for building and combining systems is gluing.

Pushout (gluing along a shared seam). A *pushout* is the specific colimit that we use constantly: given two structures A and B that share a common piece I (the *seam*), the pushout $A \oplus_I B$ is the result of gluing A and B together exactly along I and nowhere else. The defining property is *universal*: the pushout is the most economical gluing; it adds nothing and identifies nothing beyond what the seam forces.

Cocomplete (all gluings exist); complete. A category is *cocomplete* when every colimit exists inside it—you can always perform the join and stay within the same world of structures. (*Complete* is the mirror property for limits.) That **Grph** is cocomplete is what licenses us to glue agents freely and know the result is again a graph.

Functor (a map between categories). A *functor* is a structure-preserving map from one category to another: it sends objects to objects and morphisms to morphisms while respecting composition and identities. It is “a morphism between whole worlds”. A functor guarantees that the record of “do f then g ” is exactly the record of f glued to the record of g , which is precisely how histories chain together.

Topos; presheaf topos. A *topos* is a category that is rich enough to behave like the world of ordinary sets: one can form gluings and intersections, talk about subobjects (sub-structures), and reason logically inside it. A *presheaf topos* is one built from a simple template by “indexed collections of sets”, and the category of graphs is exactly such a topos.

Adhesive category (gluings are well-behaved). A category is *adhesive* when

gluing along an inclusion behaves as cleanly as it does for ordinary sets: when you glue A and B along a seam I that sits inside each as a sub-structure, the copies of A and B inside the result stay intact and overlap in exactly I —no extra merging, no surprises.

Slice category and typing. Fixing a reference object T (a fixed *type graph*, our rulebook of sorts), the *slice category* $\mathbf{Grph}/T$ has as its objects the graphs that come equipped with a *typing*—a morphism $\tau : G \rightarrow T$ labelling every node and edge with its kind (this node is an **Agent**, that edge is a *delegate*). A morphism here is a graph morphism that respects the labels. Working in the slice is simply “working with typed structures,” i.e. structures that carry their ontology with them.

From the above, the proofs invoke only three concrete facts about typed graphs, and the reader can take them as given. (1) **Gluings exist** (cocompleteness): any two typed structures can be joined along a shared seam. (2) **Gluing is computed piece-by-piece**: the joined structure is just A and B overlapping exactly on the seam, with no new nodes or edges. (3) **Gluing keeps things distinct** (adhesivity): the copies of A and B inside the result do not secretly merge.

Appendix B

Extended Formalization of the Framework

Proposition 1 (Boundary monotonicity). *If $P(a, a')$ then every external connection of a is inherited by a' : $P(x, x') \wedge C(x, y) \Rightarrow C(x', y)$.*

Proof. Immediate from the monotonicity axiom: $P(x, x')$ and $C(x, y)$ give $C(x', y)$. A connection at a finer part is inherited by the coarser part, so boundaries are preserved upward under containment. □

Theorem 1 (Composability). *Let $A, B \in \text{Str}(O)$ and let $A \leftarrow I \hookrightarrow B$ be an interface. Then (a) the gluing $Q := A \oplus_I B$ exists and carries a canonical typing $\tau_Q : Q \rightarrow T_O$; (b) $A, B \models \Phi_{\text{loc}} \Rightarrow Q \models \Phi_{\text{loc}}$ unconditionally; (c) if the gluing is core-guarded then $Q \models O$. Hence well-formed structures are closed under core-guarded gluing.*

Proof. (a) **Grph** is cocomplete and $\mathbf{Grph}/T_O \rightarrow \mathbf{Grph}$ creates colimits, so Q is the pushout computed in **Grph** with the unique mediating typing induced by τ_A, τ_B (which agree on I).

(b)–(c) The legs $I \hookrightarrow A, B$ are monos; by adhesivity the coprojections j_A, j_B are monos, the pushout creates no new edges, and Q is the union of the images of A and B overlapping exactly in I . Suppose $Q \not\models O$: some $\neg P \in \Phi_O$ fails via $h : P \rightarrow Q$ with P connected.

Case 1: $h(P)$ meets no seam vertex. Every edge of Q comes from A or B , and an A -edge and a B -edge meet only at a seam vertex; a connected image avoiding the seam lies wholly in $j_A(A)$ or $j_B(B)$, so composing with the inverse mono gives $P \rightarrow A$ or $P \rightarrow B$, contradicting $A \models \neg P$ or $B \models \neg P$. This proves (b), since Φ_{loc} -patterns use no core type and cannot reach a seam vertex.

Case 2: $h(P)$ meets a seam vertex. Then P uses a core relation, so $\neg P \in \Phi_{\text{core}}$, which the core-guard verifies on the seam, so no such h exists. Both cases are impossible, hence $Q \models O$. □

Corollary 1. *Well-formed structures with core-guarded gluings form a symmetric monoidal category under $\oplus$: composition is associative and unital (as pushouts are). Modular reasoning about composition, decomposition, and orchestration is the universal property of pushouts, independent of implementation-specific logic.*

Theorem 2 (Interoperability). *Let $\{\kappa_i : K \hookrightarrow O_i\}$ be a finite family of core embeddings (monos). The wide pushout $O_\star = \text{colim}_i O_i$ over K exists, with legs $\mu_i : O_i \rightarrow O_\star$ satisfying $\mu_i \circ \kappa_i = \mu_j \circ \kappa_j =: \mu_K$, and each μ_i is a mono. Consequently (i) every O_i embeds faithfully into $O_\star$ (no semantic collapse); (ii) any K -typed message m has one interpretation identified across all parties: $\mu_i(m_i) = \mu_K(m)$.*

Proof. **Grph** is a cocomplete topos, hence adhesive; the wide pushout of monos out of K exists by cocompleteness, built by iterated pushouts. In an adhesive category monos are stable under pushout and a pushout leg along a mono is a mono; by induction each μ_i is a mono, giving (i). The equalities $\mu_i \circ \kappa_i = \mu_K$ are the cocone conditions. For (ii), a K -typed $m \rightarrow K$ composes with κ_i then μ_i to land in $O_\star$; the cocone equation forces $\mu_i(m_i) = \mu_j(m_j) = \mu_K(m)$. $\square$

Theorem 3 (Provenance preservation). *P is a functor ($P(\text{id}) = \text{id}$, $P(g \circ f) = P(g) \circ P(f)$), and it preserves acyclicity: if $P(f), P(g)$ are DAGs and f, g compose, so is $P(g \circ f)$. Hence traceability is total—every entity has a directed path back to source entities and to the agents responsible for its generating activities.*

Proof. On a concatenation P emits the disjoint union of the two step-sequences with the shared cut entities identified—exactly composition in **Prov**—and on the empty run the discrete graph; concatenation is associative with the empty run as unit, so P is functorial, and the coprojection exhibits $P(f)$ as a subgraph of $P(g \circ f)$. For acyclicity, take topological orders $\prec_f, \prec_g$. The cut entities are outputs of f (maximal in $\prec_f$) and inputs of g (minimal in $\prec_g$). Concatenating the orders—all of $P(f)$ below the non-cut nodes of $P(g)$, cut nodes at the junction—keeps every f -edge in the lower block and every g -edge in the

upper block; cross-block edges pass only through cut nodes at the boundary. The concatenation is a topological order of $P(g \circ f)$, which is therefore acyclic. Totality is then existence of a backward path to an in-degree-zero node in a finite DAG. $\square$

Theorem 4 (Auditability). (i) *The auditable properties form a Boolean subalgebra (closed under $\neg, \wedge, \vee$): exactly the α -saturated properties.* (ii) *If $M = A \parallel B$ and Ext restricts along the coprojections of Theorems 1 and 3, then for ψ_A, ψ_B auditable, the property $r \mapsto \psi_A(r|_A) \wedge \psi_B(r|_B)$ is auditable for M .*

Proof. (i) Fibres of α partition Runs; unions of fibres are closed under complement, union, intersection and contain $\top, \perp$, and $\psi = \bar{\psi} \circ \alpha$ exactly when ψ is constant on fibres.

(ii) Compatibility gives restrictions ρ_A, ρ_B with $\alpha_A(r|_A) = \rho_A(\alpha_M(r))$, so $\psi_A(r|_A) = \bar{\psi}_A(\rho_A(\alpha_M(r)))$ factors through α_M ; the conjunction does too, hence is α_M -saturated. $\square$

Theorem 5 (Sound compositional verification). *If $M_1 \models \langle \text{Assumption}_1, \text{Guarantee}_1 \rangle$, $M_2 \models \langle \text{Assumption}_2, \text{Guarantee}_2 \rangle$, and (D1) $\pi(\text{Guarantee}_2) \subseteq \text{Assumption}_1$, (D2) $\pi(\text{Guarantee}_1) \subseteq \text{Assumption}_2$, (D3) $\text{Guarantee}_1 \wedge \text{Guarantee}_2 \subseteq \varphi$, then $M_1 \parallel M_2 \models \varphi$. Satisfaction depends only on T_M , so verifiability is independent of model-internal opacity, provided T_M soundly over-approximates real behaviour.*

Proof. For any finite trace σ of $M_1 \parallel M_2$, induct on $|\sigma|$ to show $\pi_i(\sigma) \in \text{Guarantee}_i$. Base: $\varepsilon \in \text{Guarantee}_i$. Step: assume the claim for all proper prefixes; by (D2) each proper prefix of $\pi_2(\sigma)$ lies in Assumption_2 , so $M_2 \models \langle \text{Assumption}_2, \text{Guarantee}_2 \rangle$ gives $\pi_2(\sigma) \in \text{Guarantee}_2$; symmetrically by (D1), $\pi_1(\sigma) \in \text{Guarantee}_1$. No circularity, since each step uses only strictly shorter prefixes. Then (D3) gives $\sigma \in \varphi$. Since $\models$ ranges over T_M only, $\text{Runs}_{\text{real}} \subseteq \text{Traces}(T_M)$ transfers the guarantee to real behaviour. $\square$